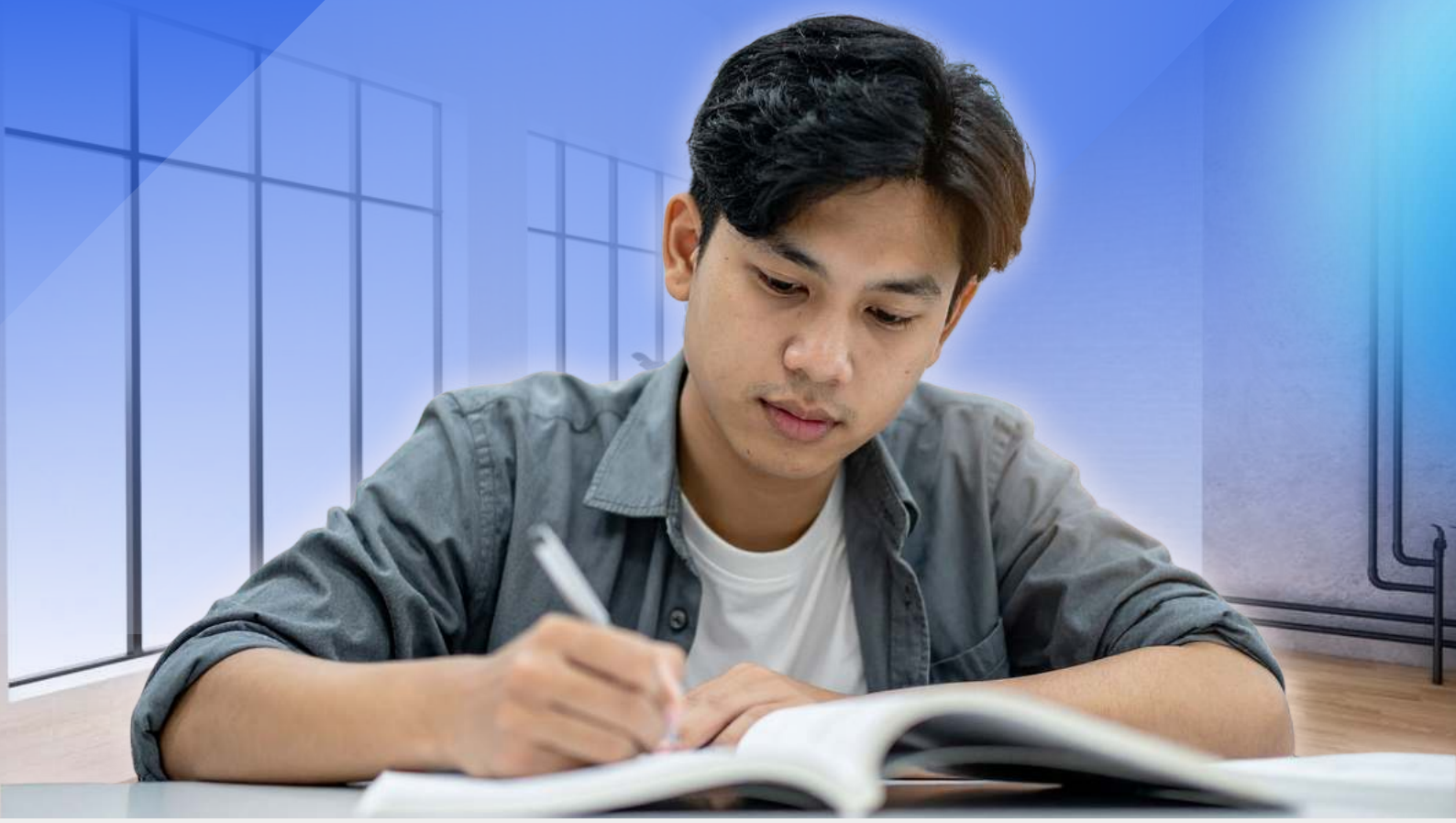
Lenterago

POST TEST

SAT MATH

61005

44 QUESTIONS



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MODUL 1

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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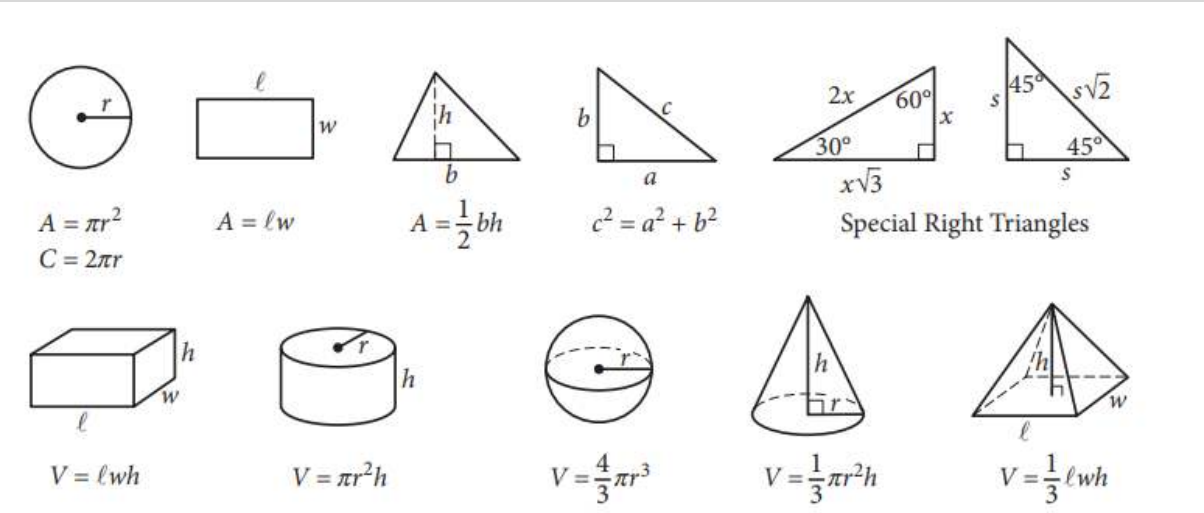

The number of degrees of arc in a circle is 360.

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The sum of the measures in degrees of the angles of a triangle is 180.

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Module 1

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1. If $x = 6$, what is the value of $x + 23$?
   A) 17
   B) 23
   C) 28
   D) 29

2. Consider two data sets.
   Data set A: 4, 8, 10, 12, 16
   Data set B: 4, 8, 10, 12
   Which statement correctly compares their means?
   A) The mean of A is greater than the mean of B.
   B) The mean of A is less than the mean of B.
   C) The means of A and B are equal.
   D) There is not enough information to compare.

3. Two right triangles ABC and DEF are similar, with right angles at C and F. Vertex A corresponds to vertex D. If $\angle B = 35°$, what is the measure of $\angle D$?
   A) 35°
   B) 45°
   C) 55°
   D) 145°

4. A launch vehicle had 520,000 kilograms of propellant before ignition. After 16 seconds, 412,800 kilograms remained. On average, how many kilograms of propellant were burned each second?
   A) 2,680
   B) 6,700
   C) 18,200
   D) 107,200

5. The system below has solution (x, y). What is $x + y$?
   $6x = 18$
   $-5x + y = 4$
   A) 12
   B) 17
   C) 22
   D) 28

6. A trainee completes 96 total hours using x in-person workshops and y online modules, modeled by $8x + 14y = 96$. By how many hours does one online module exceed one workshop?

7. Square M has side length 8 centimeters. The perimeter of square N is 3 times the perimeter of square M. What is the side length, in centimeters, of square N?
   A) 12
   B) 16
   C) 24
   D) 32

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8. A car's fuel model is $g(m) = -0.08m + 15.7$, where $g(m)$ is the gallons of gas left after driving m miles. About how many gallons are used per mile?
   A) 0.08
   B) 15.7
   C) 23
   D) 196.25

9. Fill-in: If $|5x - 7| = 143$, what is the positive value of $x - 10$?

10. The positive numbers b, x, and y satisfy $1/(9b) = 5x/y$. Which equation gives x in terms of b and y?
    A) $x = 45by$
    B) $x = y/(45b)$
    C) $x = y/(5b)$
    D) $x = 5yb$

11. Fill-in: The table shows a linear relation $f(x) = m\,x - 14$.

x	8	12	16	20
f(x)	74	118	162	206

    What is the value of m?

12. $(7x^3 - 5) - (-3x^3 + 11)$ is equivalent to $b\,x^3 - 16$. What is b?
    A) 4
    B) 8
    C) 10
    D) 16

13. Explanation: Combine: $7x^3 - 5 + 3x^3 - 11 = 10x^3 - 16 \Rightarrow b = 10$.
    The system is $y < -6$ and $3x + y < -25$. The point $(x, -7)$ is a solution. Which could be the value of x?
    A) −9
    B) −6
    C) −4
    D) 1

14. A culture starts with 250,000 cells per milliliter and triples every 4 hours. How many cells per milliliter are there after 12 hours?
    A) 750,000
    B) 2,250,000
    C) 6,750,000
    D) 27,000,000

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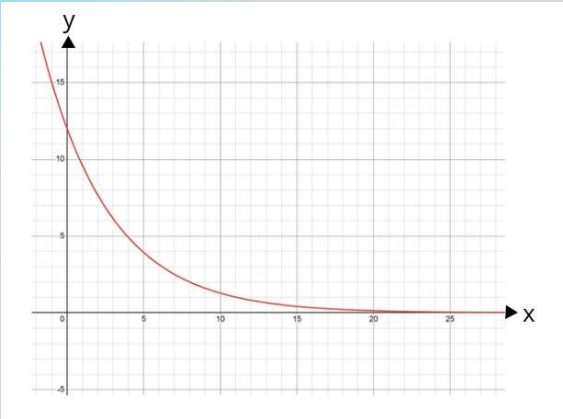
15. 

A company's graph shows the estimated number of coupons y (in thousands) mailed to customers at the end of each year, where x is the number of years since the end of 2014 ($0 \le x \le 10$). The y-intercept of the graph is 12. Which statement best interprets this y-intercept?
A) The estimated total number of coupons mailed during the first 10 years was 12,000.
B) The estimated number of coupons mailed at the end of 2014 was 12.
C) The estimated number of coupons mailed at the end of 2014 was 12,000.
D) The estimated number of coupons mailed at the end of 2024 was 12,000.

16. The population of Town B increased by 12% from 2018 to 2019. If the 2019 population is k times the 2018 population, what is k?
A) 0.12
B) 0.88
C) 1.12
D) 12

17. Which system of linear equations has no solution?
A) $y = -3x + 4$
$y = -3x - 1$
B) $x = 2$
$y = -5$
C) $y = 4x - 7$
$y = -x - 2$
D) $y = 2x + 3$
$4y = 8x + 12$

18. Which expression is equivalent to $\sqrt[3]{(x^5y^5)}$, where x and y are positive?
A) $\sqrt[5]{((xy)^3)}$
B) $\sqrt[3]{((xy)^5)}$
C) $(xy)^8$
D) $(xy)^{15}$

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19. The first term of a sequence is 8. Each subsequent term is 5 times the previous term. If $w$is the n-th term, which formula gives $w$in terms of n?

    A) $w = 5 \cdot 8^{n}$

    B) $w = 8 \cdot 5^{n-1}$

    C) $w = 8^{n} \cdot 5$

    D) $w = 5^{n} \cdot 8$

20. The minimum possible value of $x$is 9 less than 7 times another number p. Which inequality shows the possible values of x?

    A) $x \leq 7p - 9$

    B) $x \geq 7p - 9$

    C) $x \leq 9 - 7p$

    D) $x \geq 9 - 7p$

21. In right triangle RST, the side lengths are $RS = 24$, $ST = 7$, and $TR = 25$. Triangle $RST$is similar to triangle $UVW$with $S \leftrightarrow V$and $T \leftrightarrow W$. What is $\tan W$?

    A) $7/24$

    B) $24/7$

    C) $25/7$

    D) $7/25$

22. Fill-in: The line $7x - 5y = 20$is translated down 3 units in the xy-plane. What is the x-coordinate of the x-intercept of the translated line?

    Answer: 5

    Explanation: Translating down 3 makes the new x-intercept occur where the original line has $y = 3$. Plugging into $7x - 5y = 20$: $7x - 15 = 20 \Rightarrow 7x = 35 \Rightarrow x = 5$.

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MODUL 2

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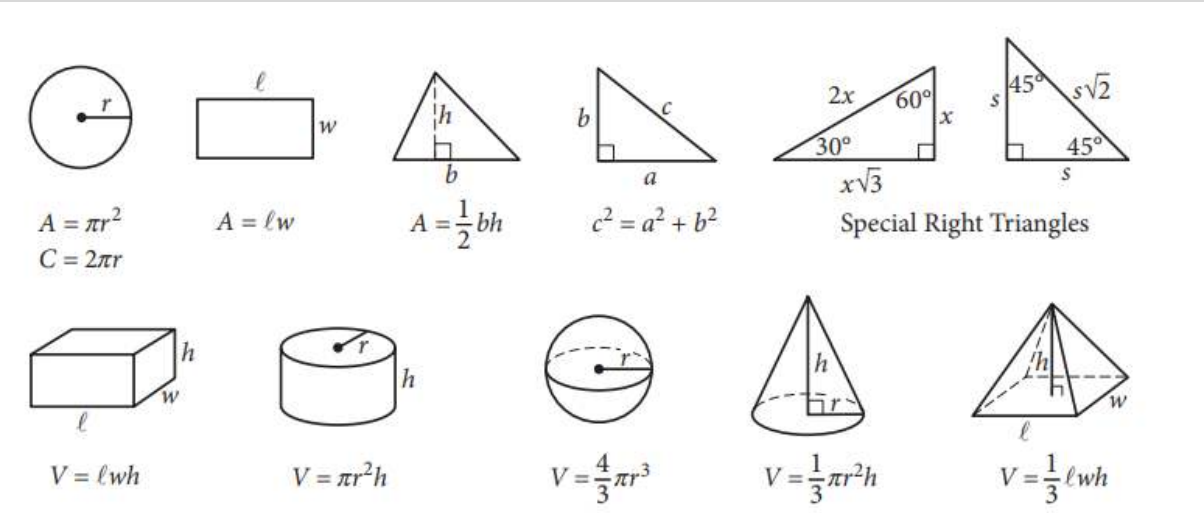

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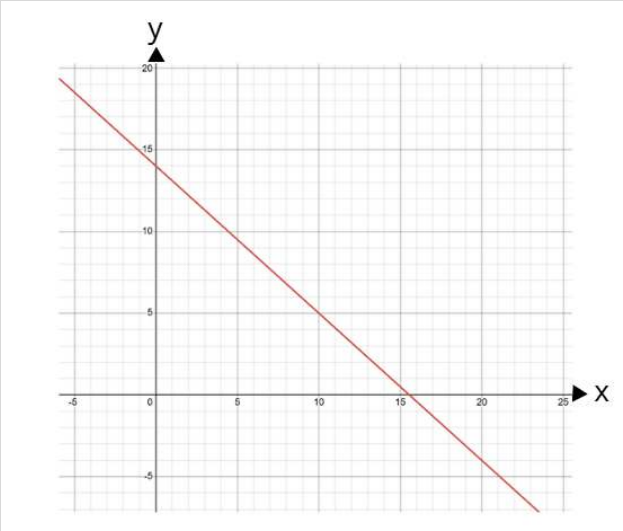
23. A language club has 72 members. A random sample is surveyed about joining a new exchange program, and 25% indicate they plan to join. Based on this, what is the best estimate of the total number of members who plan to join?
    A) 12
    B) 18
    C) 25
    D) 36

24. A machine can make large or small boxes, one at a time, for a total of 840 minutes per day. It takes 12 minutes for a large box and 6 minutes for a small box. If x is the number of large boxes and y is the number of small boxes, which equation represents the possible combinations?
    A) $6x + 12y = 840$
    B) $12x + 6y = 840$
    C) $(x + y)(12 + 6) = 840$
    D) $12x + y = 840$

25. 

    The graph shows points with a line of best fit drawn. Which equation best represents the line shown?
    A) $y = 14 + 0.9x$
    B) $y = 14 - 0.9x$
    C) $y = -14 + 0.9x$
    D) $y = -14 - 0.9x$

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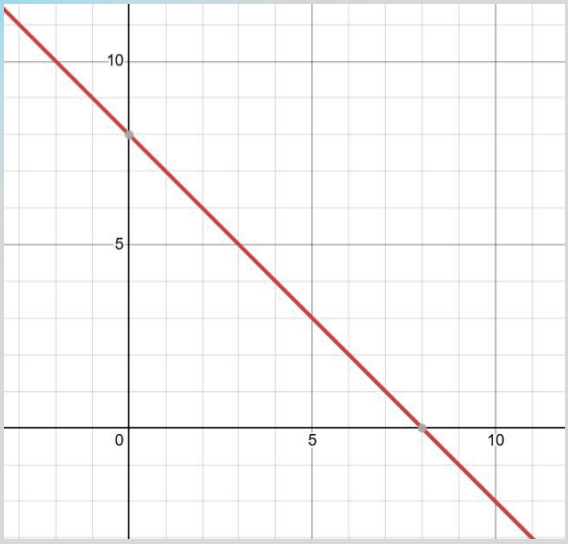
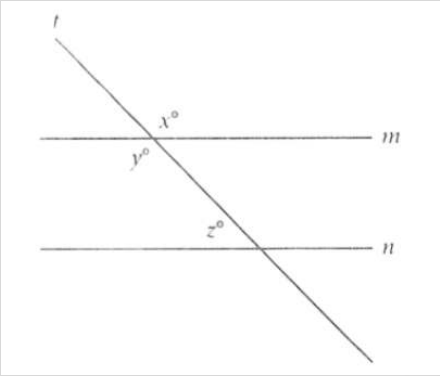
26. 

The graph shows a line passing through (0, 8) and (8, 0). The point (d, 5) lies on the line. What is the value of d?
A) 2
B) 3
C) 4
D) 5

27. For a quadratic function f, the table shows values:
x: −1 0 1
f(x): 13 12 17
Which equation defines f?
A) $f(x) = 3x^2 + 2x + 12$
B) $f(x) = 2x^2 + 3x + 12$
C) $f(x) = -x^2 + 5x + 12$
D) $f(x) = 4x^2 + 12$

28. A function is defined by $f(x) = (x + 18)/6$, and $f(a) = 9$. What is the value of a?
A) 12
B) 18
C) 36
D) 54

29. 

In the figure, lines m and n are parallel.
If $x = 5k + 20$and $y = 7k + 40$, what is the value of z?

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30. Line $p$is given by $3y - 12x = 18$. Line $r$is perpendicular to $p$in the xy-plane. What is the slope of line r?
    A) -4
    B) $-\frac{1}{4}$
    C) $\frac{1}{4}$
    D) 4

31. A wood sample has density 780kg/m³ and is a cube with edge 0.85m. To the nearest kilogram, what is the mass of the sample?
    A) 462
    B) 479
    C) 510
    D) 540

32. A city's population is modeled by $P(t) = 450\left(1.05\right)^{2t}$,where $t$is years after 2020. According to the model, the population increases by n%every 6 months. What is n?
    A) 2.5
    B) 5
    C) 10
    D) 12

33. The expression $4x^2 + bx - 45$can be written as $(hx + k)(x + j)$,where $h, k, j$are integers. Which of the following must be an integer?
    A) $\frac{b}{h}$
    B) $\frac{b}{k}$
    C) $\frac{45}{h}$
    D) $\frac{45}{k}$

34. Fill-in: For what value of $k$does the equation $\left(k - 1\right)x = 2k - 3$have no solution for x?

35. Fill-in: A shop buys an item for \$8. The retail price is 250%of wholesale. At season's end, items are discounted 70% off the retail price. What is the discounted price (in dollars) per item?

36. Fill-in: The quadratic $x^2 - 22x + c = 0$has no real solutions if $c > n$. What is the least possible value of n?

37. Fill-in: Set A has 60 buildings, mean height 28m. Set B has 90 buildings, mean height 52m. Set C combines all 150 buildings. What is the mean height of set C, in meters?

38. A system is given by
    $y = -2$and $y = x^2 + 6x + a$,where $a > 0$.
    The system has exactly one distinct real solution. What is the value of a?

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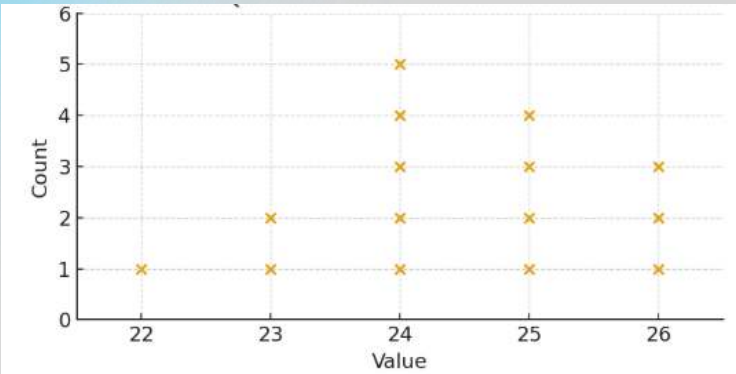
39. 

A dot plot shows 15 values for data set A (somewhere around 22–26).
Data set B is created by adding 40 to every value of A. Which statement correctly compares medians and ranges of A and B?
A) Medians equal; ranges equal.
B) Medians equal; range(B) greater.
C) Median(B) greater; ranges equal.
D) Median(B) greater; range(B) greater.

40. Consider $45x^2 + (45b + a)x + ab = 0$,where $a, b > 0$. The product of the solutions equals $kab$for some constant k. What is k?
A) $\frac{1}{9}$
B) $\frac{1}{15}$
C) $\frac{1}{45}$
D) 45

41. The graph of $x^2 + x + y^2 + y = \frac{449}{2}$is a circle. What is its radius?

42. Two identical right rectangular prisms have height 135cm and square base of side s. Each prism's surface area is Kcm². Gluing them together along a square base forms a taller prism. The new surface area is $\frac{92}{47}K$cm². What is s?
A) 6
B) 8
C) 12
D) 18

43. A parabola has vertex $(7, -12)$and crosses the x-axis at two points. If it is written $y = ax^2 + bx + c$,which could be the value of $a + b + c$?
A) -20
B) -15
C) -12
D) -9

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44. A rational function is defined by $f(x) = \frac{a}{x+b}$,where $a$and $b$are constants.

It is known that $f$has a vertical asymptote at $x = -4$and $f(0) = \frac{3}{2}$.

Define $g(x) = f(x+4)$ .Which equation could define g?

A) $g(x) = \frac{6}{x}$

B) $g(x) = \frac{6}{x+4}$

C) $g(x) = \frac{6}{x+8}$

D) $g(x) = \frac{6x}{x+4}$

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